



Time: 5 minutes

Outcome: Apply Profit-Maximizing Output & Monopoly Price.

Difficulty: Intermediate

Profit-Maximizing Output & Monopoly Price

THE CORE IDEA

A monopolist compares marginal revenue with marginal cost to choose output, then reads the corresponding price from demand. In the displayed interior solution, MR exceeds MC before their crossing and falls below MC afterward. The output decision and the price decision are separate; marginal cost need not always slope upward.

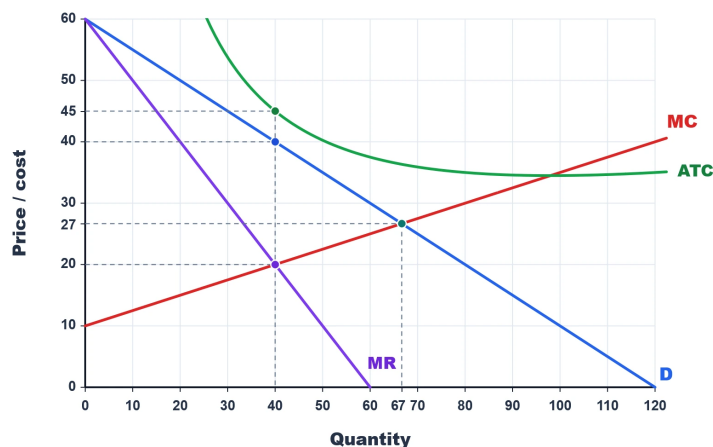
HOW TO RECOGNIZE IT

- Find the output where MR crosses MC from above in the interior solution.
- Move vertically from that output to the demand curve to find the monopoly price.
- Keep the output decision and price decision separate: marginal conditions determine quantity, demand determines price.
- Do not set price equal to marginal revenue or marginal cost unless the graph happens to imply it.

WATCH OUT

A common mistake is to read monopoly price from the marginal-revenue curve. Find output where $MR = MC$, then read the price buyers will pay from demand.

WORKED EXAMPLE: FINDING MONOPOLY PRICE



The monopolist first finds output where $MR = MC$, which occurs at 40 units and a marginal value of \$20. It then moves up to the demand curve, where buyers will pay \$40 for 40 units. The profit-maximizing choice is therefore $Q = 40$ and $P = \$40$, not the \$20 MR value.

CHECK YOURSELF

What is the correct order for solving a monopoly graph?